MASTER'S THESIS PROPOSAL

Cross-Cultural Multimodal Hate Speech Detection for Roman Script South Asian Content Using Vision-Language Models

1. Introduction

Social media has changed how people communicate, and unfortunately, also how they harm each other. Across South Asia, hundreds of millions of Pakistani and Indian users write Urdu and Hindi in Roman script rather than their official scripts (Rizwan, Shakeel, & Karim, 2020; Ashiq, Kanwal, Rafique, & Waqas, 2024). This is largely a practical choice: standard Urdu (Nastaliq) and Hindi (Devanagari) are difficult to type on a regular keyboard or phone, so Roman script became the default for everyday digital communication (Ashiq, Kanwal, Rafique, & Waqas, 2024). This shift has created a problem that has gone largely unaddressed. Roman Urdu and Roman Hindi now form a shared digital script space that official language boundaries do not reflect, and content written in it moves freely across national borders without needing to change. Within this space, hateful content that combines images with Roman script text, largely in the form of memes and image-caption posts, has spread at scale, causing real harm to communities defined by religion, ethnicity, gender, and political identity. Analysis of 22.4 million social media posts in India, combined with records of attacks on religious minorities between 2020 and 2022, found that online religious-nationalist content was associated with a measurable increase in real-world attacks on Muslim and Christian communities (Schutte, Karell, & Barrett, 2025). Because Pakistan and India share both the script and much of the cultural reference pool, this content may follow patterns of harm that have never actually been studied (Bui, von der Wense, & Lauscher, 2025).

This research looks at two distinct types of harmful multimodal content. The first is memes, where Roman script text is embedded directly inside the image. The second is image-caption posts, where an image is paired with a separate Roman Urdu or Roman Hindi caption. In both cases, the meaning comes from the combination of image and text together, and in both cases, a system that only reads one side of that combination will miss the harm sitting in the other. A system that only reads the text of a meme misses the image. A system that only reads the image misses the caption. Neither is enough on its own. Only a system that reads both together, and understands the cultural context behind them, can catch this kind of harm reliably.

Most existing hate speech detection systems are built for English. The few that address Urdu or Hindi focus only on text (Rizwan, Shakeel, & Karim, 2020; Ashiq, Kanwal, Rafique, & Waqas, 2024), and none of them combine text with images, which means they miss the meme and the image-caption post entirely, since meaning in those formats comes from combining image and text, not from either alone. Even the strongest multilingual multimodal hate speech datasets available today, such as Multi3Hate (Bui, von der Wense, & Lauscher, 2025), which covers five languages including Hindi script, contain no Roman Urdu or Roman Hindi content at all. The South Asian digital space is simply not served by any of this work, whether the system reads text alone or combines both modalities.

This research sets out to close that gap: a multimodal hate speech detection framework for Roman script South Asian content, built to handle both memes and image-caption posts, using vision-language models fine-tuned on a purpose-built annotated dataset, the Roman Script Multimodal Hate Speech Dataset (RSMHSD), collected from Pakistani and Indian social media. To the best of our knowledge, RSMHSD is the first dataset to combine both image and text for Roman Urdu and Roman Hindi hate speech content, moving beyond the text-only resources that currently exist for this language space.

2. Background and Motivation

2.1 The Roman Script Problem

Roman Urdu and Roman Hindi are written in the same Latin script and share a lot of vocabulary, grammar, and online slang (Rizwan, Shakeel, & Karim, 2020; Ashiq, Kanwal, Rafique, & Waqas, 2024). When a Pakistani writes "yaar ye bilkul ghalat hai" and an Indian writes "yaar ye bilkul galat hai," most NLP models cannot even tell the difference. That overlap raises an interesting question: could a system trained on Pakistani social media content actually generalise to Indian content, and the other way around? Nobody has tested this at the multimodal level yet. This research does not assume an answer either way; it tests it directly and reports the first empirical evidence either for or against it.

2.2 Why Multimodal South Asian Content Is Different

South Asian online hate does not stay confined to text. This research focuses on two distinct types of content that together make up the multimodal hate speech problem in this space. The first is the meme, where Roman script text sits embedded directly inside the image, and meaning comes from the specific pairing of the visual and the overlaid text. The second is the image-caption post, where a standalone image is paired with a separate Roman Urdu or Roman Hindi caption typed underneath it. Structurally, these are different problems. Memes need OCR to pull out the embedded text before any linguistic analysis can happen, while image-caption posts hand over text and image as two already-separate inputs. But both share the same core issue: the hateful intent lives across both modalities at once, and neither one alone tells the full story (Bui, von der Wense, &

Lauscher, 2025; Saddozai, Badri, & Alghazzawi, 2025). Traditional text-only NLP pipelines simply cannot handle either case. This research is built to handle both.

2.3 Connection to Prior Work

This research builds directly on my final year project, a cyberbullying detection system that reached 92% accuracy on 40,000+ English social media comments using a Random Forest classifier with TF-IDF features. That system had a clear limitation. It performed well in English but could not touch the Roman Urdu content where South Asian communities actually experience harm day to day. This Master's research picks up exactly where that left off, moving from unimodal English text classification into multimodal, cross-cultural South Asian content.

3. Research Gaps

A review of the current literature reveals five specific gaps that this research addresses.

Gap 1: No multimodal dataset exists for Roman script South Asian content, neither memes nor image-caption posts

Existing Roman Urdu hate speech datasets available on GitHub, Kaggle, and Hugging Face are exclusively text-only (Rizwan, Shakeel, & Karim, 2020; Ashiq, Kanwal, Rafique, & Waqas, 2024). No publicly available dataset contains annotated multimodal samples, neither memes with embedded Roman script text nor image-caption posts with Roman Urdu or Roman Hindi captions, covering South Asian hate speech. Both content types are entirely absent from existing resources. This is the single largest gap in the field, and building the first such dataset is the primary contribution of this research.

Gap 2: Existing multimodal datasets exclude South Asian languages

The most advanced multilingual multimodal hate speech dataset, Multi3Hate, covers English, German, Spanish, Hindi script, and Mandarin, but not Roman Urdu or Roman Hindi (Bui, von der Wense, & Lauscher, 2025). Roman script South Asian content is completely absent from every major multimodal benchmark currently available.

Gap 3: No vision-language model has been fine-tuned for this specific task

Vision-language models such as CLIP, BLIP-2, and LLaVA have shown strong multimodal understanding across a range of classification tasks. Yet existing multimodal hate speech systems for Urdu still rely on older architectures such as BiLSTM combined with EfficientNetB1 (Saddozai, Badri, & Alghazzawi, 2025), and major multilingual multimodal benchmarks such as Multi3Hate contain no Roman Urdu or Roman Hindi content at all (Bui, von der Wense, & Lauscher, 2025). To the best of my knowledge, no prior work has fine-tuned vision-language models specifically for Roman script South

Asian hate speech detection, leaving this capability completely unaddressed in the literature.

Gap 4: Cross-cultural South Asian meme context is unaddressed

Pakistani and Indian online spaces share substantial overlap in Roman script usage, internet slang, meme formats, and references to shared historical events, including Partition, cricket rivalry, and Bollywood and Lollywood crossover culture. This overlap means meme content frequently circulates across both national audiences without modification. At the same time, the two countries diverge sharply in majority religion, political narratives, and cultural sensitivities, meaning the same content can carry opposite meaning depending on the reader's background. Cross-cultural annotation studies in other multilingual contexts have shown that cultural background produces systematic disagreement in hate speech labeling (Bui et al., 2025), and there is reason to expect a similar effect for Pakistani and Indian annotators given the depth of their cultural and political divergence. Yet to the best of our knowledge, no existing study has examined this cross-cultural dynamic for Roman script South Asian content, neither the degree of overlap in meme formats and hate speech patterns, nor the degree of divergence in cultural interpretation. This research is the first to test it empirically.

Gap 5: Sarcasm and irony in South Asian memes

Sarcasm, irony, and culturally coded humour are well-documented challenges for automated hate speech detection systems (Waseem, Davidson, & Warmsley, 2017; Davidson, Warmsley, Macy, & Weber, 2017). A classifier trained on explicit hate speech patterns will systematically misclassify content where hateful intent is communicated indirectly, whether through a sarcastic caption that means the opposite of what it says, or through a culturally loaded image that only carries meaning for an in-group audience. South Asian meme culture leans heavily on these indirect forms. References to Partition trauma, religious symbolism, regional stereotypes, and cricket rivalry are routinely weaponized in ways that look benign to an outsider but are immediately recognizable as hateful to the intended audience. While implicit hate detection has been studied for English content (Caselli, Basile, Mitrović, & Granitzer, 2021), to the best of our knowledge no existing system addresses implicit hate expressed through sarcasm and cultural reference in Roman script South Asian multimodal content. This research takes a first step toward that challenge, by constructing a dataset annotated by culturally aware Pakistani and Indian annotators who can recognize indirect hate, and by using vision-language models that process image and text jointly, which offers richer context than text-only systems for detecting hate communicated across both modalities at once. Fully resolving sarcasm and implicit hate detection remains an open challenge and is identified here as a direction for future work.

4. Research Objectives

This research has four primary objectives:

Construct a purpose-built multimodal hate speech dataset (RSMHSD, the Roman Script Multimodal Hate Speech Dataset) comprising Pakistani and Indian memes and image-caption posts, annotated with binary hate speech labels (Hateful / Not Hateful) and cultural origin (Pakistani / Indian).
Fine-tune vision-language models (CLIP and LLaVA) on the RSMHSD dataset to perform joint image-text hate speech classification for Roman script South Asian content.
Benchmark the proposed system against unimodal baselines (text-only transformer, image-only CNN) and existing multimodal approaches, to evaluate the relative contribution of joint visual-linguistic reasoning for this task.
Conduct a cross-cultural analysis examining how Pakistani and Indian annotators perceive the same multimodal content differently, and how this cultural variation affects model performance.

5. Research Questions

RQ1. Do vision-language models fine-tuned on Roman script South Asian content outperform text-only and image-only baselines for multimodal hate speech detection, and does incorporating visual signals improve overall detection performance on content that relies on sarcasm or culturally coded reference, identified through qualitative error analysis on misclassified samples rather than a dedicated implicit/explicit hate label?

RQ2. What is the degree of cross-cultural variation in hate speech annotation between Pakistani and Indian annotators evaluating the same multimodal content, and does this variation systematically affect model performance across cultural subsets?

RQ3. Can a single unified model handle both Pakistani and Indian meme and image-caption content effectively, or does cross-cultural variation require separate fine-tuning?

6. Dataset Strategy

6.1 Existing Datasets (Starting Point)

The following existing datasets will be used as supplementary resources and baselines:

Dataset	Size	Type	Limitation
Roman Urdu Hate Speech (GitHub/Kaggle)	~10K samples	Text only	No images or memes
MMHS150K (Gomez et al., 2020)	150,000 image-text pairs	Multimodal	No Roman script
Multi3Hate (Bui et al., 2025)	Multilingual benchmark	Multimodal	No Urdu/Roman Hindi
Hateful Memes (Kiela et al., 2020)	10,000 memes	Multimodal	English only
HuggingFace Roman Urdu HS	Large corpus	Text only	No visual content

6.2 New Dataset Construction (Primary Contribution)

The core dataset contribution of this thesis is the Roman Script Multimodal Hate Speech Dataset (RSMHSD). Construction follows four stages.

Stage 1: Data Collection

Multimodal content will be collected through systematic manual curation from publicly accessible pages and accounts on Twitter/X, Facebook, and Instagram. Given current API restrictions across these platforms, collection will rely on manual browsing and downloading of publicly visible posts rather than programmatic scraping, in full compliance with each platform's terms of service. Content will be sourced from prominent Pakistani and Indian public pages, regional meme accounts, and politically and socially active accounts, using a predefined set of Roman Urdu and Roman Hindi seed keywords covering four harm categories: religion, ethnicity, gender, and political discourse. To ensure cross-cultural balance, Pakistani-origin and Indian-origin content will be tracked and collected separately throughout this stage.

The target corpus is 6,000 image-text pairs, a scale comparable to established multimodal hate speech benchmarks including Hateful Memes (10,000 samples), while remaining feasible to collect and annotate within the first six months of the project timeline. The dataset will be constructed with a balanced class structure (3,000 hateful, 3,000 not hateful) to ensure stable classifier training and fair cross-cultural comparison, a standard design choice in hate speech dataset construction (Bui, von der Wense, & Lauscher, 2025; Saddozai, Badri, & Alghazzawi, 2025). Each class will contain exactly 1,500 Pakistani and 1,500 Indian samples to guarantee equal cultural representation across both training and evaluation. Within each cultural subset, samples are equally divided between the two content types, 750 memes and 750 image-caption posts per cultural origin per class, yielding a fully balanced 2×2×2 design across content type, cultural origin, and hate speech label. This structured balance ensures that no content type, cultural origin, or label class dominates the dataset, enabling fair cross-cultural and cross-modality comparison throughout all experimental evaluations.

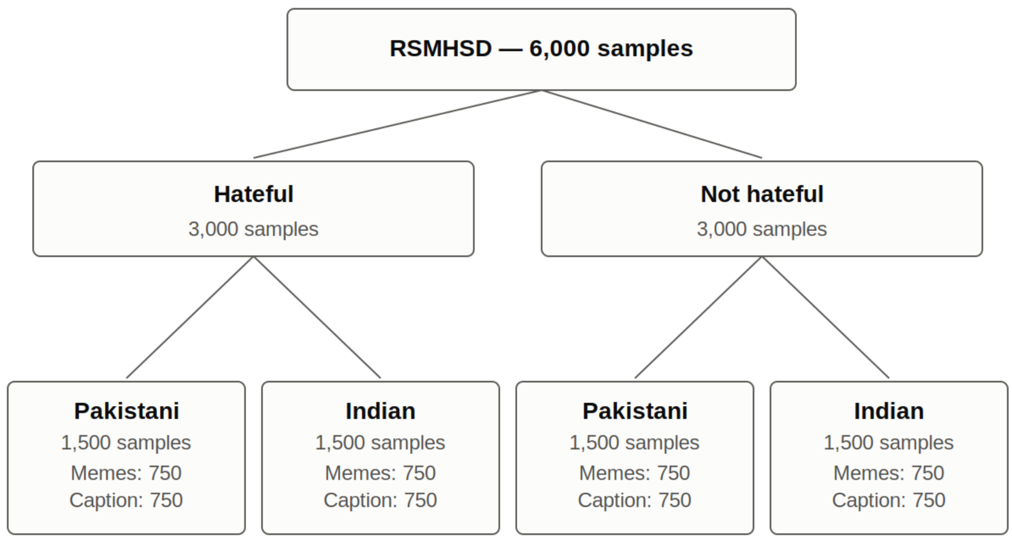


Figure 1: RSMHSD dataset composition — a balanced 2×2×2 design across label, cultural origin, and content type

Stage 2: Annotation

Each sample will be independently evaluated by a panel of three annotators: one Pakistani bilingual annotator, one Indian bilingual annotator, and one tie-breaker annotator with documented exposure to both Pakistani and Indian cultural and linguistic contexts (for example, diaspora background); this annotator is not claimed to be culturally neutral, but serves to break 2-1 splits. Annotators will be recruited through university networks and research community contacts across Pakistan and India, with a modest honorarium offered to ensure participation feasibility and acknowledge the demanding nature of hate speech annotation work.

Given the demanding nature of hate speech annotation, annotators will not be required to label more than a fixed daily quota, and will have access to breaks between annotation sessions. A brief debrief will be offered at the end of the annotation period, and annotators may withdraw at any point without affecting their honorarium for work already completed. Pakistani and Indian annotators will be recruited and coordinated independently through separate university and research-community networks in each country; no direct annotator-to-annotator communication is required, which limits exposure to cross-border logistical or political friction during recruitment.

Annotators will classify each sample using a binary labeling framework, Hateful or Not Hateful, following the standard established by comparable multimodal hate speech datasets including Hateful Memes and Multi3Hate (Bui, von der Wense, & Lauscher, 2025). Binary classification is chosen over multi-class labeling to maximize annotator agreement reliability and ensure sufficient samples per class for model training. Each annotation decision must be based on the combined context of the image and the Roman script text together. Annotation guidelines will explicitly prohibit single-modality labeling and will include worked examples of meme and image-caption samples where meaning only emerges from both modalities.

Anonymized demographic data, including cultural and regional background, will be recorded for each annotator to support the cross-cultural disagreement analysis in Research Objective 4.

Stage 3: Quality Control

Inter-annotator agreement will be measured using Fleiss' Kappa across the three-annotator panel. Samples will be retained using a majority vote threshold (2/3 agreement), the standard approach in hate speech annotation (Bui, von der Wense, & Lauscher, 2025), rather than requiring unanimous consensus, which would risk discarding a methodologically valuable proportion of genuinely ambiguous samples. Cases where Pakistani and Indian annotators disagree but the tie-breaker annotator sides with one will be retained and flagged as culturally contested samples, forming a dedicated analysis subset for Research Objective 4. A global Fleiss' Kappa score will be reported on the finalized dataset.

The dataset will be segmented into Pakistani and Indian cultural subsets throughout collection and finalized here, enabling out-of-domain cross-cultural evaluation experiments assessing how localized slang and imagery affect model generalizability.

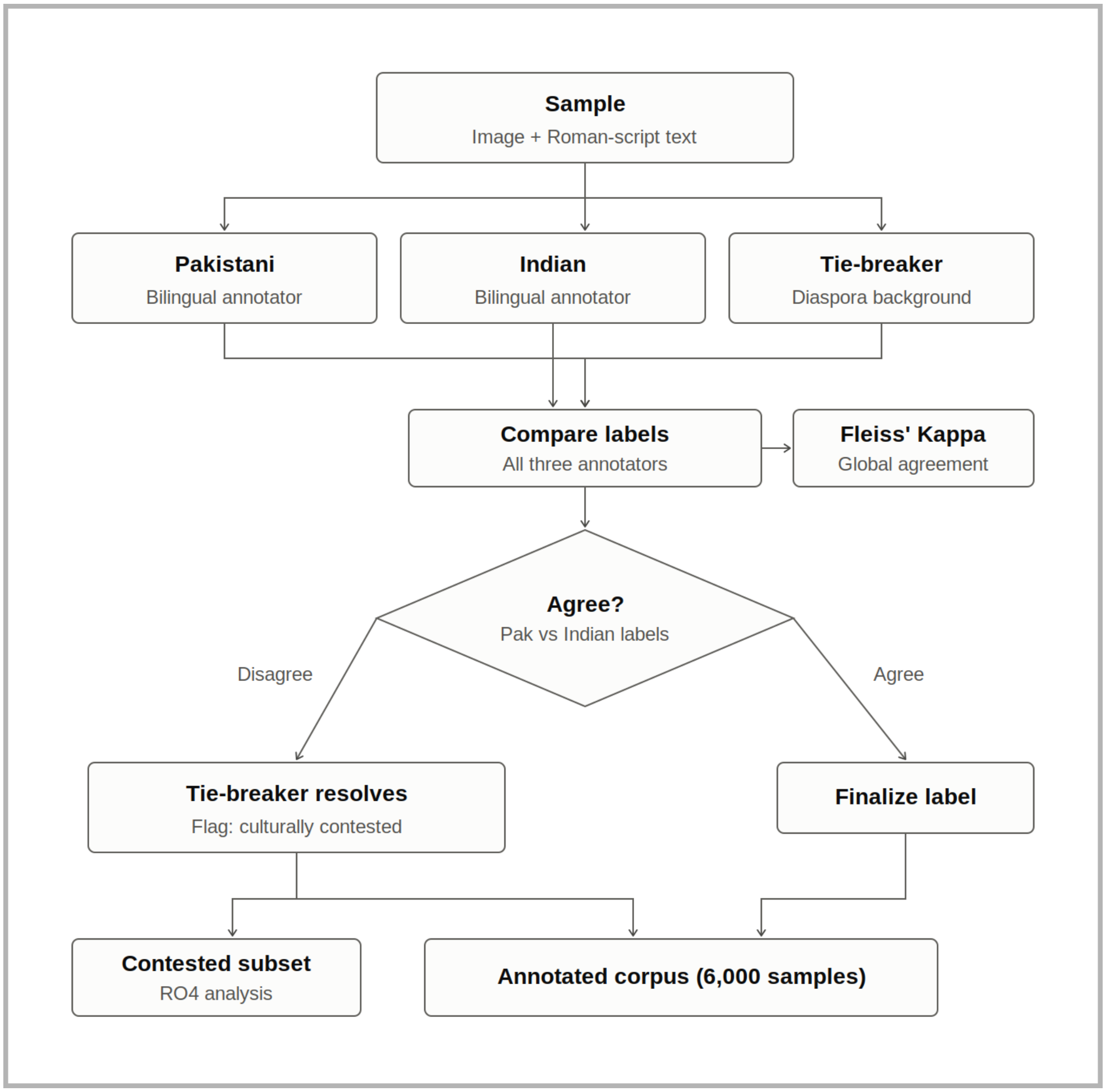


Figure 2: Three-annotator labeling and quality-control workflow

Stage 4: Public Release

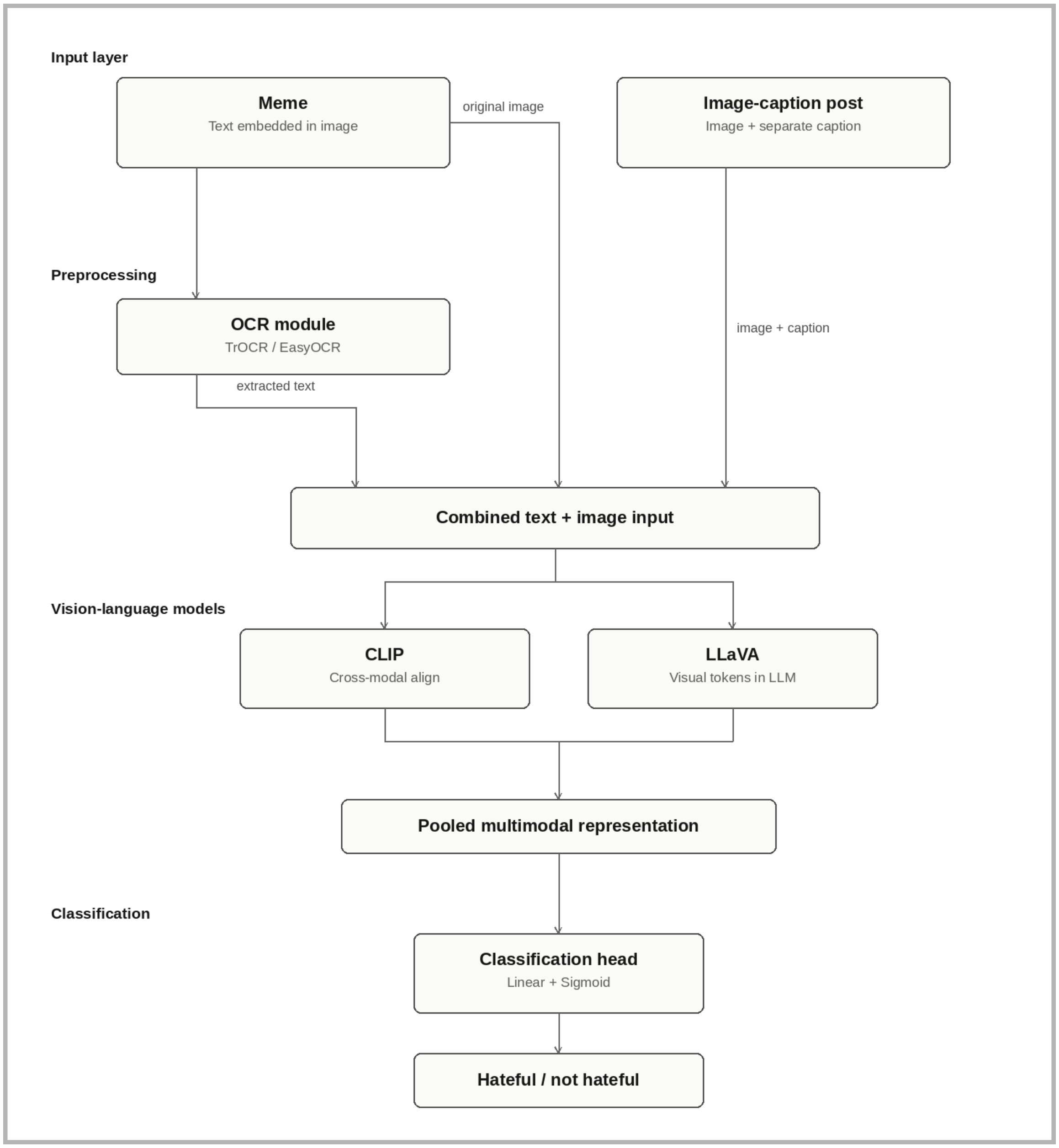
RSMHSD will be released publicly on Hugging Face and GitHub under a Creative Commons CC-BY-NC 4.0 license, permitting academic use and redistribution while prohibiting commercial applications, an appropriate restriction for a dataset containing potentially harmful content. Where full image redistribution is not legally permissible due to platform terms of service or copyright restrictions on meme content, metadata files, annotation labels, and retrieval scripts will be released in place of raw images, a standard approach adopted by datasets including Hateful Memes and MMHS150K.

7. Methodology

7.1 System Architecture

The proposed pipeline handles two input types through parallel preprocessing branches. For meme inputs, where Roman script text is embedded inside the image, an OCR module first extracts the embedded text before both the extracted text and the original image are passed to the vision-language model. TrOCR and EasyOCR will be evaluated for this extraction step, with final selection based on character error rate (CER) measured on a held-out sample of at least 500 South Asian memes representing the decorative fonts, overlays, and mixed-script text typical of this content. For image-caption posts, where an image appears alongside a separately typed Roman script caption, text and image are passed directly as separate modality inputs with no OCR step required.

Once both modalities are available as inputs, they are combined through the fusion mechanism native to each vision-language model. CLIP uses contrastive cross-modal alignment, and LLaVA concatenates visual tokens directly into the language model input sequence. For LLaVA specifically, the OCR step will be evaluated as optional, since LLaVA can directly read text embedded in images, and the end-to-end pipeline will be compared against the OCR-assisted pipeline to determine which yields better performance on meme inputs.



[Figure 3: System architecture for multimodal hate speech classification]

7.2 Annotation Protocol

The full annotation protocol, including annotator panel composition, binary labeling framework, modality-combined labeling requirement, majority vote threshold, culturally contested sample handling, and Fleiss' Kappa measurement, is described in detail in Section 6.2, Stages 2 and 3. The methodology here focuses on model development and evaluation.

7.3 Vision-Language Models

Parameter-efficient fine-tuning via LoRA (Low-Rank Adaptation) will be used to adapt large vision-language models under realistic compute constraints, an approach shown to match full fine-tuning performance on classification tasks (Hu et al., 2021). Pre-trained weights will be loaded from Hugging Face Hub using the standard checkpoints for each model. A binary classification head (Linear layer, Sigmoid) will be added on top of each model's pooled multimodal representation.

LoRA fine-tuning will be conducted on department GPU cluster resources or cloud compute credits secured ahead of Stage 2 of the methodology, using mixed-precision training to reduce memory footprint. CLIP and the BiLSTM + EfficientNet-B4 baseline are feasible on a single consumer-grade GPU (≤16 GB VRAM); LLaVA (7B) requires higher-memory GPUs (≥24 GB VRAM) or gradient checkpointing. Confirmed access to this compute will be secured before model development begins. Should compute access prove limited, the model set will be further scaled back to prioritise LLaVA as the primary proposed architecture.

Two vision-language models will be fine-tuned on RSMHSD:

- CLIP (openai/clip-vit-large-patch14): contrastive cross-modal alignment between image patches and text tokens, with the pooled joint embedding passed to the classification head. Fine-tuned with LoRA applied to the text and vision encoders. CLIP serves as a lightweight, standard VLM baseline.
- LLaVA (llava-hf/llava-1.5-7b-hf): an instruction-tuned VLM that concatenates visual tokens directly into the LLM input sequence, allowing it to reason jointly over image and text using its multimodal instruction-following capabilities. LoRA fine-tuning is applied to the language model layers. Since LLaVA is instruction-tuned for open-ended generation rather than discriminative classification, it will also be evaluated in a zero-shot prompting setting (asking the model directly whether a sample is hateful) alongside the fine-tuned classification-head setup, to check whether the added classification head is actually needed for this task.

BLIP-2 was scoped out of the primary benchmark to focus computational resources on the most promising architectures; it may be evaluated in future work if time and resources permit.

7.4 Baseline Models

Five models in total form the comparative study: three baselines and two proposed VLMs. Baselines are designed to isolate the contribution of each modality and fusion strategy.

Model	Type	Input
mBERT	Text baseline	Text only
ResNet-50	Image baseline	Image only
BiLSTM + EfficientNet-B4	Traditional fusion (Saddozai et al., 2025)	Both text and image
CLIP	Vision-language model	Both text and image
LLaVA	Vision-language model	Both text and image

mBERT receives OCR-extracted Roman script text for memes and the typed caption for image-caption posts. ResNet-50 receives visual input only, with no text signal. BiLSTM + EfficientNet-B4 replicates the existing state-of-the-art Urdu multimodal approach (Saddozai, Badri, & Alghazzawi, 2025), serving as the direct prior-work baseline. This separation ensures that any performance gains from the proposed VLM approach can be attributed specifically to joint visual-linguistic reasoning, rather than to dataset size or label distribution.

7.5 Evaluation Metrics

All models will be evaluated on a held-out test set comprising 10% of RSMHSD (600 samples), stratified by class label and cultural origin to ensure equal representation of Pakistani and Indian content in evaluation. The primary evaluation metric is macro-averaged F1-score, chosen because it treats both classes equally regardless of any residual class imbalance and is the standard metric for hate speech detection benchmarks. Supporting metrics include Accuracy, Precision, Recall (macro and weighted), and AUROC. All metrics will be reported separately for the full test set and for Pakistani and Indian cultural subsets, to enable direct cross-cultural performance comparison.

7.6 Ablation Study

A structured ablation study will isolate the contribution of each system component by comparing text-only input, image-only input, and full multimodal input, for each of the two proposed VLMs (CLIP and LLaVA). This directly addresses RQ1 by quantifying how much performance improvement is attributable specifically to the visual modality and to joint visual-linguistic reasoning. Misclassified samples from each condition will be qualitatively reviewed to check whether visual signals disproportionately help on cases involving sarcasm or culturally coded reference, since the dataset does not carry a dedicated implicit/explicit hate label.

7.7 Ethics and Data Handling

All content will be collected from publicly accessible social media pages, with no private or restricted material included. No personally identifying information will be retained in the released dataset; usernames, profile links, and other identifying metadata will be stripped during preprocessing, and only content necessary for classification will be preserved. This research will be submitted for institutional ethics review prior to data collection, covering both the handling of potentially harmful content and the recruitment of human annotators. Annotators will be informed in advance of the nature of the material and will provide consent before participating.

8. Expected Contributions

This research is designed to produce four distinct contributions to the field:

1. The Roman Script Multimodal Hate Speech Dataset (RSMHSD) itself, the first publicly available annotated dataset of South Asian multimodal hate speech content in Roman script.
2. A model benchmark, comparing five models across text-only, image-only, traditional fusion, and vision-language architectures on this dataset.
3. An empirical cross-cultural analysis of how Pakistani and Indian annotators diverge when labeling the same multimodal content, and what that divergence actually means for model performance.
4. A deployable hate speech detection pipeline, capable of handling both meme and image-caption inputs in Roman Urdu and Roman Hindi.

9. Timeline

The proposed 24-month timeline is structured across four phases:

- Months 1 to 6 focus on dataset construction: data collection, annotator recruitment and training, annotation, and quality control.
- Months 7 to 12 cover model development: the preprocessing pipeline, OCR evaluation, baseline training, and VLM fine-tuning with LoRA.
- Months 13 to 18 address evaluation and analysis: comparative benchmarking, ablation studies, cross-cultural disagreement analysis, and error analysis.

- Months 19 to 24 are dedicated to writing and dissemination: thesis writing, conference paper submission, dataset release on Hugging Face and GitHub, and final thesis submission.

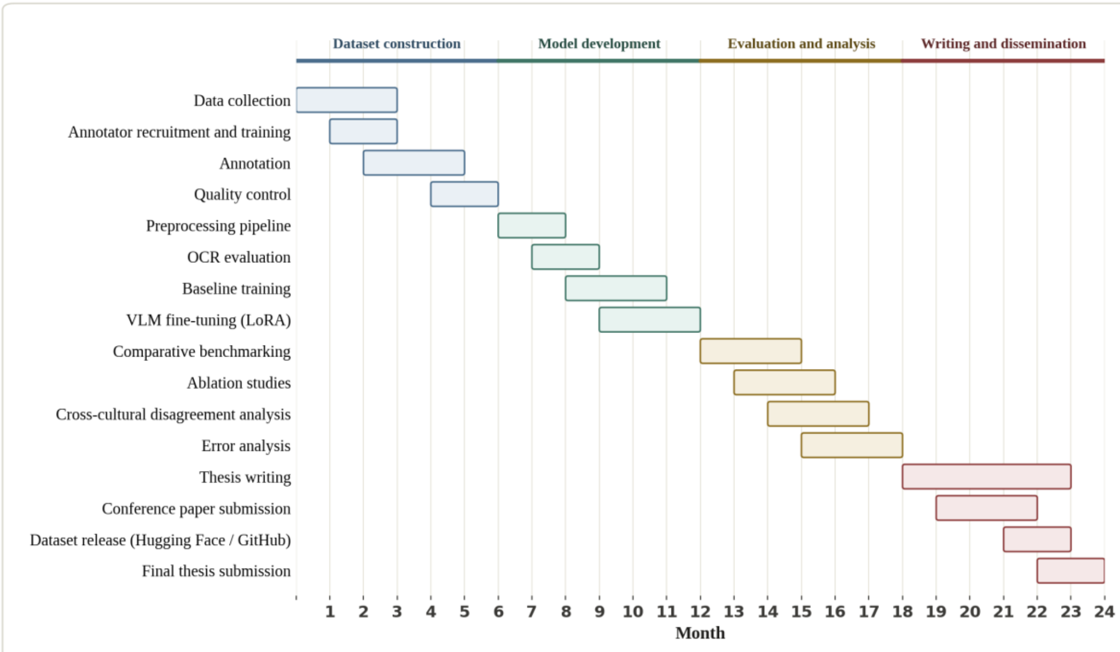


Figure 4: Proposed 24-month project timeline

10. Conclusion

Hate speech in Roman script South Asian content sits in a genuine blind spot. It is multimodal, it is culturally specific, and until now, it has had no dedicated dataset, no fine-tuned vision-language models, and no empirical study of how Pakistani and Indian communities actually experience and interpret it differently. This research addresses that gap directly, by building the first dataset of its kind, benchmarking five models to isolate what joint image-text reasoning actually contributes, and running the first cross-cultural annotation study for this specific content space.

The work is scoped to be achievable within 24 months, while still producing four outputs that each stand on their own: a public dataset of 6,000 samples, a model benchmark, a cross-cultural analysis, and a working detection pipeline. Beyond the immediate contributions, this research opens a path for extending multimodal hate speech detection to other underserved language communities, and for eventually applying the same framework to multicultural contexts beyond South Asia.

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